

Phys 115B HW 4

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1 D

Question 1. *An Electron Spin State (Griffiths 4.30, which is 4.27 in the second edition)*

An electron is in the spin state $\chi = A \begin{pmatrix} 3i \\ 4 \end{pmatrix}$.

- (a) Determine the normalization constant A .
- (b) Find the expectation values of S_x , S_y , and S_z .
- (c) Find the “uncertainties” σ_{S_x} , σ_{S_y} , and σ_{S_z} .
- (d) Confirm that your results are consistent with all three uncertainty principles.

Proof. Here we'll assume the spin state is in the basis consists of eigenvectors of S_z .

- (a) Given that the state is normalized, $1 = \|\chi\| = \sqrt{|A3i|^2 + |A4|^2} = |A|\sqrt{9+16} = 5|A|$, so $|A| = \frac{1}{5}$.
WLOG, pick A to be positive real, hence $A = \frac{1}{5}$.

- (b) First, recall that the matrix in z -basis are given as following:

$$S_x = \frac{\hbar}{2} \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}, \quad S_y = \frac{\hbar}{2} \begin{pmatrix} 0 & -i \\ i & 0 \end{pmatrix}, \quad S_z = \frac{\hbar}{2} \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix} \quad (1)$$

Hence, the expectation values are given as follow:

$$\langle S_x \rangle = \chi^\dagger S_x \chi = \frac{\hbar}{2 \cdot 25} \begin{pmatrix} -3i & 4 \end{pmatrix} \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix} \begin{pmatrix} 3i \\ 4 \end{pmatrix} = \frac{\hbar}{50} \begin{pmatrix} -3i & 4 \end{pmatrix} \begin{pmatrix} 4 \\ 3i \end{pmatrix} = 0 \quad (2)$$

$$\langle S_y \rangle = \chi^\dagger S_y \chi = \frac{\hbar}{2 \cdot 25} \begin{pmatrix} -3i & 4 \end{pmatrix} \begin{pmatrix} 0 & -i \\ i & 0 \end{pmatrix} \begin{pmatrix} 3i \\ 4 \end{pmatrix} = \frac{\hbar}{50} \begin{pmatrix} -3i & 4 \end{pmatrix} \begin{pmatrix} -4i \\ -3 \end{pmatrix} = -\frac{24\hbar}{50} = -\frac{12\hbar}{25} \quad (3)$$

$$\langle S_z \rangle = \chi^\dagger S_z \chi = \frac{\hbar}{2 \cdot 25} \begin{pmatrix} -3i & 4 \end{pmatrix} \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix} \begin{pmatrix} 3i \\ 4 \end{pmatrix} = \frac{\hbar}{50} \begin{pmatrix} -3i & 4 \end{pmatrix} \begin{pmatrix} 3i \\ -4 \end{pmatrix} = -\frac{7\hbar}{50} \quad (4)$$

- (c) For the uncertainties, we also need to compute the expectation values of the squares. For simplicity, we first compute the square of the operators:

$$S_x^2 = S_y^2 = S_z^2 = \frac{\hbar^2}{4} \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix} \quad (5)$$

Then, the expectation values become:

$$\langle S_x^2 \rangle = \langle S_y^2 \rangle = \langle S_z^2 \rangle = \chi^\dagger S_z^2 \chi = \frac{\hbar^2}{4} \chi^\dagger I_2 \chi = \frac{\hbar^2}{4} \chi^\dagger \chi = \frac{\hbar^2}{4} \|\chi\|^2 = \frac{\hbar^2}{4} \quad (6)$$

Then, the corresponding uncertainties are:

$$\sigma_{S_x} = \sqrt{\langle S_x^2 \rangle - \langle S_x \rangle^2} = \sqrt{\frac{\hbar^2}{4} - 0^2} = \frac{\hbar}{2} \quad (7)$$

$$\sigma_{S_y} = \sqrt{\langle S_y^2 \rangle - \langle S_y \rangle^2} = \sqrt{\frac{\hbar^2}{4} - \left(\frac{-24\hbar}{50}\right)^2} = \frac{\hbar}{2} \sqrt{1 - \left(\frac{24}{25}\right)^2} = \frac{7\hbar}{25} \quad (8)$$

$$\sigma_{S_z} = \sqrt{\langle S_z^2 \rangle - \langle S_z \rangle^2} = \sqrt{\frac{\hbar^2}{4} - \left(\frac{-7\hbar}{50}\right)^2} = \frac{\hbar}{2} \sqrt{1 - \left(\frac{7}{25}\right)^2} = \frac{24\hbar}{25} \quad (9)$$

- (d) If compute the product of any two uncertainties, we get the following:

$$\sigma_{S_x} \sigma_{S_z} = \frac{7\hbar^2}{50}, \quad \sigma_{S_x} \sigma_{S_z} = \frac{24\hbar^2}{50} = \frac{12\hbar^2}{25}, \quad \sigma_{S_y} \sigma_{S_z} = \frac{168\hbar^2}{625} \quad (10)$$

Which, using the expectation values deduced in part (b), we have the following:

$$\frac{1}{2} |\langle [S_x, S_y] \rangle| = \frac{1}{2} |\langle i\hbar S_z \rangle| = \frac{\hbar}{2} |\langle S_z \rangle| = \frac{7\hbar^2}{100} \quad (11)$$

$$\frac{1}{2} |\langle [S_y, S_z] \rangle| = \frac{1}{2} |\langle i\hbar S_x \rangle| = \frac{\hbar}{2} |\langle S_x \rangle| = 0 \quad (12)$$

$$\frac{1}{2} |\langle [S_z, S_x] \rangle| = \frac{1}{2} |\langle i\hbar S_y \rangle| = \frac{\hbar}{2} |\langle S_y \rangle| = \frac{6\hbar^2}{25} \quad (13)$$

Which, one results in the following:

$$\sigma_{S_x} \sigma_{S_y} = \frac{7\hbar^2}{50} \geq \frac{7\hbar^2}{100} = \frac{1}{2} |\langle [S_x, S_y] \rangle| \quad (14)$$

$$\sigma_{S_x} \sigma_{S_z} = \frac{12\hbar^2}{25} \geq \frac{6\hbar^2}{25} = \frac{1}{2} |\langle [S_z, S_x] \rangle| \quad (15)$$

$$\sigma_{S_y} \sigma_{S_z} = \frac{168\hbar^2}{625} \geq 0 = \frac{1}{2} |\langle [S_y, S_z] \rangle| \quad (16)$$

So, the uncertainty principles are preserved.

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2 D

Question 2. Spin in $\hat{y}$ (Griffiths 4.32, 4.29 in second edition)

I wrote down the answers in lecture - derive the answers the same way we did it for S_x in class.

- (a) Find the eigenvalues and eigenspinors of S_y .
- (b) If you measured S_y on a particle in the general state $\chi = a\chi_+ + b\chi_-$, what values might you get, and what is the probability of each? Check that the probabilities add up to 1.
- (c) If you measured S_y^2 , what values might you get, and with what probabilities?

Proof. First, if using the z -basis, then the raising and lowering operators have the following actions:

$$S_+\chi_+ = \mathbf{0}, \quad S_+\chi_- = \hbar\chi_+, \quad S_-\chi_+ = \hbar\chi_-, \quad S_-\chi_- = \mathbf{0} \quad (17)$$

And, by definition $S_{\pm} = S_x \pm iS_y$. So, one also gets $S_+ - S_- = 2iS_y$, or $S_y = -\frac{i}{2}(S_+ - S_-)$. Hence, the actions on the basis elements are given as follow:

$$S_y\chi_+ = -\frac{i}{2}(-\hbar\chi_-) = i\frac{\hbar}{2}\chi_-, \quad S_y\chi_- = -\frac{i}{2}(\hbar\chi_+) = -i\frac{\hbar}{2}\chi_+ \quad (18)$$

Hence, the matrices are written as follow under the z -basis:

$$\mathcal{M}(S_y) = \frac{\hbar}{2} \begin{pmatrix} 0 & -i \\ i & 0 \end{pmatrix} \quad (19)$$

- (a) Taking the characteristic polynomial of the matrix $A = \begin{pmatrix} 0 & -i \\ i & 0 \end{pmatrix}$, one gets the following:

$$\det(A - \lambda I) = \det \begin{pmatrix} -\lambda & -i \\ i & -\lambda \end{pmatrix} = \lambda^2 - 1 \quad (20)$$

Hence, $\lambda^2 - 1 = 0$ implies the eigenvalues are ± 1 . Which, solving the equations about the eigenvectors of $\lambda = 1$, one gets the following:

$$\lambda = 1, \quad v_1 = \begin{pmatrix} a \\ b \end{pmatrix} \in \ker(A - I) \implies \begin{pmatrix} -1 & -i \\ i & -1 \end{pmatrix} \begin{pmatrix} a \\ b \end{pmatrix} = \begin{pmatrix} -a - ib \\ ia - b \end{pmatrix} = \mathbf{0} \quad (21)$$

Which, it implies that $a = -ib$, and $b = ia$. WLOG, take $a = \frac{1}{\sqrt{2}}$, then one has $b = \frac{i}{\sqrt{2}}$, showing $v_1 = \frac{1}{\sqrt{2}} \begin{pmatrix} 1 \\ i \end{pmatrix}$.

Now, solving the equations about the eigenvectors of $\lambda = -1$, one gets the following:

$$\lambda = -1, \quad v_2 = \begin{pmatrix} a \\ b \end{pmatrix} \in \ker(A + I) \implies \begin{pmatrix} 1 & -i \\ i & 1 \end{pmatrix} \begin{pmatrix} a \\ b \end{pmatrix} = \begin{pmatrix} a - ib \\ ia + b \end{pmatrix} = \mathbf{0} \quad (22)$$

Which, it implies that $a = ib$, and $b = -ia$. WLOG, take $a = \frac{1}{\sqrt{2}}$, then one has $b = -\frac{i}{\sqrt{2}}$, showing $v_2 = \frac{1}{\sqrt{2}} \begin{pmatrix} 1 \\ -i \end{pmatrix}$.

As $S_y = \frac{\hbar}{2}A$, then it inherits all eigenvectors, with corresponding eigenvalues rescaled by factor of $\frac{\hbar}{2}$. Hence, we get:

$$\lambda = \frac{\hbar}{2}, \chi_+^{(y)} = v_1 = \frac{1}{\sqrt{2}} \begin{pmatrix} 1 \\ i \end{pmatrix} = \frac{\chi_+ + i\chi_-}{\sqrt{2}} \quad (23)$$

$$\lambda = -\frac{\hbar}{2}, \chi_-^{(y)} = v_2 = \frac{1}{\sqrt{2}} \begin{pmatrix} 1 \\ -i \end{pmatrix} = \frac{\chi_+ - i\chi_-}{\sqrt{2}} \quad (24)$$

(b) Given $\chi_+^{(y)} = \frac{1}{\sqrt{2}}(\chi_+ + i\chi_-)$ and $\chi_-^{(y)} = \frac{1}{\sqrt{2}}(\chi_+ - i\chi_-)$, one gets that $\chi_+^{(y)} + \chi_-^{(y)} = \frac{2}{\sqrt{2}}\chi_+ = \sqrt{2}\chi_+$, so $\chi_+ = \frac{1}{\sqrt{2}}(\chi_+^{(y)} + \chi_-^{(y)})$. On the other hand, one also gets $\chi_+^{(y)} - \chi_-^{(y)} = \frac{2i}{\sqrt{2}}\chi_- = i\sqrt{2}\chi_-$, so one has $\chi_- = -\frac{i}{\sqrt{2}}(\chi_+^{(y)} - \chi_-^{(y)})$.

Hence, given the state $\chi = a\chi_+ + b\chi_-$ with $\|\chi\|^2 = |a|^2 + |b|^2 = 1$, one has the following in terms of the eigenvectors of S_y :

$$\chi = a\chi_+ + b\chi_- = \frac{a}{\sqrt{2}}(\chi_+^{(y)} + \chi_-^{(y)}) + \frac{bi}{\sqrt{2}}(\chi_+^{(y)} - \chi_-^{(y)}) = \frac{a+bi}{\sqrt{2}}\chi_+^{(y)} + \frac{a-bi}{\sqrt{2}}\chi_-^{(y)} \quad (25)$$

Hence, with $a = a_u + ia_v$ and $b = b_u + ib_v$ for $a_u, a_v, b_u, b_v \in \mathbb{R}$ (where the condition $|a|^2 + |b|^2 = 1$ translates to $a_u^2 + a_v^2 + b_u^2 + b_v^2 = 1$), one has the following:

$$\chi = \frac{(a_u - b_v) + i(a_v + b_u)}{\sqrt{2}}\chi_+^{(y)} + \frac{(a_u + b_v) + i(a_v - b_u)}{\sqrt{2}}\chi_-^{(y)} \quad (26)$$

Thus, with the observables of S_y being $\pm\frac{\hbar}{2}$, the above states have the following probabilities of measuring each observables:

$$\mathbb{P}\left(\lambda = \frac{\hbar}{2}\right) = \left| \frac{(a_u - b_v) + i(a_v + b_u)}{\sqrt{2}} \right|^2 = \frac{1}{2}((a_u - b_v)^2 + (a_v + b_u)^2) = \frac{1 + 2(a_v b_u - a_u b_v)}{2} \quad (27)$$

$$\mathbb{P}\left(\lambda = -\frac{\hbar}{2}\right) = \left| \frac{(a_u + b_v) + i(a_v - b_u)}{\sqrt{2}} \right|^2 = \frac{1}{2}((a_u + b_v)^2 + (a_v - b_u)^2) = \frac{1 - 2(a_v b_u - a_u b_v)}{2} \quad (28)$$

These two values do add up to 1.

Then, notice that $a\bar{b} = (a_u + ia_v)(b_u - ib_v) = (a_u b_u + a_v b_v) + i(a_v b_u - a_u b_v)$, so the term $(a_v b_u - a_u b_v) = \text{Im}(a\bar{b})$. Hence, the probabilities can also be written as:

$$\mathbb{P}\left(\lambda = \frac{\hbar}{2}\right) = \frac{1 + 2 \cdot \text{Im}(a\bar{b})}{2}, \quad \mathbb{P}\left(\lambda = -\frac{\hbar}{2}\right) = \frac{1 - 2 \cdot \text{Im}(a\bar{b})}{2} \quad (29)$$

Where $\bar{b}$ is the complex conjugate of b .

(c) If measuring S_y^2 , notice that its eigenvectors are the same as S_y , with eigenvalues being the square $\frac{\hbar^2}{4}$ (for both $\pm\frac{\hbar}{2}$). Hence, no matter which state it collapses to in the y -basis, we'll always measure $\frac{\hbar^2}{4}$. There is a total probability of 1 to measure $\frac{\hbar^2}{4}$.

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Question 3. *Spin One (Griffiths 4.34, 4.31 in second edition)*

This is very similar to what we did in the matrix representation lecture, but simpler, as there is no total $s = 0$ case to worry about (spin is fixed and immutable).

Construct the spin matrices S_x, S_y , and S_z for a particle of spin 1.

Hint: How many eigenstates of S_z are there? Determine the action of S_z, S_+ , and S_- on each of these states. Follow the procedure used in the text for spin $\frac{1}{2}$.

Proof.

I. Deduce the scaling of $S_{\pm}$

First, for spin $s = 1$, we know the allowed number m is $m \in \{-1, 0, 1\}$. Here, we'll use $\chi_{-1}, \chi_0, \chi_1$ to denote the three states corresponding to $m = -1, 0, 1$ as eigenvectors of S_z (with eigenvalues $-\hbar, 0, \hbar$ respectively). Hence, one has the following equations:

$$S^2\chi_m = \hbar^2 s(s+1)\chi_m = 2\hbar^2\chi_m, \quad S_z\chi_m = \hbar m\chi_m \quad (30)$$

Now, consider the raising and lowering operators $S_{\pm} = S_x \pm iS_y$, so one has $S_{\pm}^{\dagger} = S_x^{\dagger} \pm (-i)S_y^{\dagger} = S_x \mp iS_y = S_{\mp}$. Using the commutation relation similar to $\mathfrak{so}_3$, one also deduces the following:

$$S_+S_- = (S_x + iS_y)(S_x - iS_y) = S_x^2 + S_y^2 - iS_xS_y + iS_yS_x = S_x^2 + S_y^2 - i[S_x, S_y] = S_x^2 + S_y^2 + \hbar S_z \quad (31)$$

$$[S_+, S_-] = [S_x + iS_y, S_x - iS_y] = -i[S_x, S_y] + i[S_y, S_x] = -2i \cdot i\hbar S_z = 2\hbar S_z \quad (32)$$

$$\implies S_-S_+ = S_+S_- - [S_+, S_-] = S_x^2 + S_y^2 - \hbar S_z \quad (33)$$

Hence, with the total spin $S^2 = S_x^2 + S_y^2 + S_z^2$, one rewrite it as follow:

$$S^2 = (S_x^2 + S_y^2 + \hbar S_z) - \hbar S_z + S_z^2 = S_+S_- - \hbar S_z + S_z^2 = S_-S_+ + \hbar S_z + S_z^2 \quad (34)$$

Then, suppose $S_{\pm}\chi_m$ are scaled from some normalized states (or possibly 0), one gets the following actions on each basis vector:

χ_1 :

$$2\hbar^2\chi_1 = S^2\chi_1 = (S_+S_- - \hbar S_z + S_z^2)\chi_1 = S_+S_-\chi_1 - \hbar^2\chi_1 + \hbar^2\chi_1 = S_+S_-\chi_1 \quad (35)$$

$$= (S_-S_+ + \hbar S_z + S_z^2)\chi_1 = S_-S_+\chi_1 + \hbar^2\chi_1 + \hbar^2\chi_1 = S_-S_+\chi_1 + 2\hbar^2\chi_1 \quad (36)$$

Which implies the following on the norm:

$$S_+S_-\chi_1 = 2\hbar^2\chi_1 \implies 2\hbar^2 = \langle \chi_1 | 2\hbar^2\chi_1 \rangle = \langle \chi_1 | S_+S_-\chi_1 \rangle = \langle S_-\chi_1 | S_-\chi_1 \rangle \implies \|S_-\chi_1\| = \sqrt{2}\hbar \quad (37)$$

$$S_-S_+\chi_1 = \mathbf{0} \implies 0 = \langle \chi_1 | S_-S_+\chi_1 \rangle = \langle S_+\chi_1 | S_+\chi_1 \rangle \implies \|S_+\chi_1\| = 0 \implies S_+\chi_1 = \mathbf{0} \quad (38)$$

χ_0 :

$$2\hbar^2\chi_0 = S^2\chi_0 = (S_+S_- - \hbar S_z + S_z^2)\chi_0 = S_+S_-\chi_0 \quad (39)$$

$$= (S_-S_+ + \hbar S_z + S_z^2)\chi_0 = S_-S_+\chi_0 \quad (40)$$

Which implies the following on the norm:

$$S_+S_-\chi_0 = 2\hbar^2\chi_0 \implies 2\hbar^2 = \langle \chi_0 | 2\hbar^2\chi_0 \rangle = \langle \chi_0 | S_+S_-\chi_0 \rangle = \langle S_-\chi_0 | S_-\chi_0 \rangle \implies \|S_-\chi_0\| = \sqrt{2}\hbar \quad (41)$$

$$S_-S_+\chi_0 = 2\hbar^2\chi_0 \implies 2\hbar^2 = \langle \chi_0 | 2\hbar^2\chi_0 \rangle = \langle \chi_0 | S_-S_+\chi_0 \rangle = \langle S_+\chi_0 | S_+\chi_0 \rangle \implies \|S_+\chi_0\| = \sqrt{2}\hbar \quad (42)$$

χ_{-1} :

$$2\hbar^2\chi_{-1} = S^2\chi_{-1} = (S_+S_- - \hbar S_z + S_z^2)\chi_{-1} = S_+S_-\chi_{-1} + 2\hbar^2\chi_{-1} \quad (43)$$

$$= (S_-S_+ + \hbar S_z + S_z^2)\chi_{-1} = S_-S_+\chi_{-1} - \hbar^2\chi_{-1} + \hbar^2\chi_{-1} = S_-S_+\chi_{-1} \quad (44)$$

Which implies the following on the norm:

$$S_+S_-\chi_{-1} = \mathbf{0} \implies 0 = \langle \chi_{-1} | S_+S_-\chi_{-1} \rangle = \langle S_-\chi_{-1} | S_-\chi_{-1} \rangle \implies \|S_-\chi_{-1}\| = 0 \implies S_-\chi_{-1} = \mathbf{0} \quad (45)$$

$$S_-S_+\chi_{-1} = 2\hbar^2\chi_{-1} \implies 2\hbar^2 = \langle \chi_{-1} | 2\hbar^2\chi_{-1} \rangle = \langle \chi_{-1} | S_-S_+\chi_{-1} \rangle = \langle S_+\chi_{-1} | S_+\chi_{-1} \rangle \implies \|S_+\chi_{-1}\| = \sqrt{2}\hbar \quad (46)$$

II. Deduce the vector $S_{\pm}\chi_m$

First, let's consider the commutation relation of S_z with $S_{\pm}$:

$$[S_z, S_{\pm}] = [S_z, S_x] \pm i[S_z, S_y] = i\hbar S_y \mp i \cdot i\hbar S_x = i\hbar S_y \pm \hbar S_x = \pm\hbar(S_x \pm iS_y) = \pm\hbar S_{\pm} \quad (47)$$

So, the action of S_z on the above $S_{\pm}\chi_m$ (that're not zero vectors) are as follow:

$$S_z(S_-\chi_1) = ([S_z, S_-] + S_-S_z)\chi_1 = -\hbar S_-\chi_1 + \hbar S_-\chi_1 = 0(S_-\chi_1) \implies S_-\chi_1 \in \text{span}\{\chi_0\} \quad (48)$$

$$S_z(S_+\chi_0) = ([S_z, S_+] + S_+S_z)\chi_0 = \hbar S_+\chi_0 + \mathbf{0} = \hbar(S_+\chi_0) \implies S_+\chi_0 \in \text{span}\{\chi_1\} \quad (49)$$

$$= S_z(S_-\chi_0) = ([S_z, S_-] + S_-S_z)\chi_0 = -\hbar S_-\chi_0 + \mathbf{0} = -\hbar(S_-\chi_0) \implies S_-\chi_0 \in \text{span}\{\chi_{-1}\} \quad (50)$$

$$= S_z(S_+\chi_{-1}) = ([S_z, S_+] + S_+S_z)\chi_{-1} = \hbar S_+\chi_{-1} - \hbar S_+\chi_{-1} = 0(S_+\chi_{-1}) \implies S_+\chi_{-1} \in \text{span}\{\chi_0\} \quad (51)$$

Combine with the norm deduced in the previous section, one gets:

$$S_+\chi_1 = \mathbf{0}, \quad S_-\chi_1 = \sqrt{2}\hbar\chi_0, \quad S_+\chi_0 = \sqrt{2}\hbar\chi_1, \quad S_-\chi_0 = \sqrt{2}\hbar\chi_{-1}, \quad S_+\chi_{-1} = \sqrt{2}\hbar\chi_0, \quad S_-\chi_{-1} = \mathbf{0} \quad (52)$$

Hence, representing S_+, S_- as matrices under the basis $\chi_1, \chi_0, \chi_{-1}$, one has the following:

$$\mathcal{M}(S_+) = \begin{pmatrix} 0 & \sqrt{2}\hbar & 0 \\ 0 & 0 & \sqrt{2}\hbar \\ 0 & 0 & 0 \end{pmatrix} = \sqrt{2}\hbar \begin{pmatrix} 0 & 1 & 0 \\ 0 & 0 & 1 \\ 0 & 0 & 0 \end{pmatrix}, \quad \mathcal{M}(S_-) = \begin{pmatrix} 0 & 0 & 0 \\ \sqrt{2}\hbar & 0 & 0 \\ 0 & \sqrt{2}\hbar & 0 \end{pmatrix} = \sqrt{2}\hbar \begin{pmatrix} 0 & 0 & 0 \\ 1 & 0 & 0 \\ 0 & 1 & 0 \end{pmatrix} \quad (53)$$

III. Spin matrices under z -basis

Since $S_{\pm} = S_x \pm iS_y$, one has $S_+ + S_- = 2S_x$, so $S_x = \frac{1}{2}(S_+ + S_-)$; on the other hand, $S_+ - S_- = 2iS_y$, so $S_y = \frac{-i}{2}(S_+ - S_-)$. Which, the corresponding spin matrices are as follow (with the fact that $\chi_1, \chi_0, \chi_{-1}$ are eigenvectors of S_z with $\hbar, 0, -\hbar$ as their eigenvalues):

$$\mathcal{M}(S_x) = \frac{1}{2}(\mathcal{M}(S_+) + \mathcal{M}(S_-)) = \frac{\hbar}{\sqrt{2}} \begin{pmatrix} 0 & 1 & 0 \\ 1 & 0 & 1 \\ 0 & 1 & 0 \end{pmatrix} \quad (54)$$

$$\mathcal{M}(S_y) = -\frac{i}{2}(\mathcal{M}(S_+) - \mathcal{M}(S_-)) = \frac{\hbar}{\sqrt{2}} \begin{pmatrix} 0 & -i & 0 \\ i & 0 & -i \\ 0 & i & 0 \end{pmatrix} \quad (55)$$

$$\mathcal{M}(S_z) = \hbar \begin{pmatrix} 1 & 0 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & -1 \end{pmatrix} \quad (56)$$

□

4 D

Question 4. *General Spin Operator (Modified Griffiths 4.33, 4.30 in second edition)*

Construct the matrix S_r corresponding to the component of spin angular momentum along an arbitrary direction $\hat{r}$. Use spherical coordinates, for which

$$\hat{r} = \sin \theta \cos \phi \hat{x} + \sin \theta \sin \phi \hat{y} + \cos \theta \hat{z}$$

- (a) Doing no math, what do we expect the eigenvalues of S_r to be?
- (b) Calculate the 2×2 matrix $S_r = \hat{r} \cdot \mathbf{S}$.
- (c) Find the eigenvectors and eigenvalues of S_r explicitly. (The answer is in the book)
- (d) We measure S_z for a spin $1/2$ particle and get $\hbar/2$. Now measure S_r . What is the probability of getting $\hbar/2$? Check that the answer makes sense for $\theta = 0$ and $\theta = \pi/2$.
- (e) The above constructed the state ψ_r which is the eigenstate with $\lambda = \hbar/2$ of an arbitrary spin projection operator S_r . Argue that the converse is always true: ANY state ψ is the eigenstate of some spin projection operator S_n . This means that if you are given a spin $1/2$ particle in any random spin state, there is some axis on which it is fully polarized (and we might as well call that axis the z axis).

Proof.

- (a) If it's for spin- $\frac{1}{2}$, the expected eigenvalues of S_r are still $\frac{\hbar}{2}, -\frac{\hbar}{2}$.

- (b) Given the definition $S_r = \hat{r} \cdot \mathbf{S}$, one has the following matrix under the fixed z -basis:

$$\mathcal{M}(S_r) = \sin(\theta) \cos(\phi) \mathcal{M}(S_x) + \sin(\theta) \sin(\phi) \mathcal{M}(S_y) + \cos(\theta) \mathcal{M}(S_z) \quad (57)$$

$$= \frac{\hbar}{2} \begin{pmatrix} \cos(\theta) & \sin(\theta)(\cos(\phi) - i \sin(\phi)) \\ \sin(\theta)(\cos(\phi) + i \sin(\phi)) & -\cos(\theta) \end{pmatrix} \quad (58)$$

$$= \frac{\hbar}{2} \begin{pmatrix} \cos(\theta) & \sin(\theta)e^{-i\phi} \\ \sin(\theta)e^{i\phi} & -\cos(\theta) \end{pmatrix} \quad (59)$$

- (c) We'll consider $\mathcal{M}(S_r) = \frac{\hbar}{2}A$ for the matrix A written above, then scale the eigenvectors / eigenvalues correspondingly.

Consider the characteristic polynomial:

$$\det(A - \lambda I) = \det \begin{pmatrix} \cos(\theta) - \lambda & \sin(\theta)e^{-i\phi} \\ \sin(\theta)e^{i\phi} & -\cos(\theta) - \lambda \end{pmatrix} = (\lambda + \cos(\theta))(\lambda - \cos(\theta)) - \sin^2(\theta) = \lambda^2 - 1 \quad (60)$$

So, the eigenvalues are $\lambda = \pm 1$. For the eigenvectors, we get:

$$\lambda = 1, \quad v_1 = \begin{pmatrix} a \\ b \end{pmatrix} \in \ker(A - I) \implies \begin{pmatrix} \cos(\theta) - 1 & \sin(\theta)e^{-i\phi} \\ \sin(\theta)e^{i\phi} & -\cos(\theta) - 1 \end{pmatrix} \begin{pmatrix} a \\ b \end{pmatrix} = \mathbf{0} \quad (61)$$

$$\lambda = -1, \quad v_{-1} = \begin{pmatrix} c \\ d \end{pmatrix} \in \ker(A + I) \implies \begin{pmatrix} \cos(\theta) + 1 & \sin(\theta)e^{-i\phi} \\ \sin(\theta)e^{i\phi} & -\cos(\theta) + 1 \end{pmatrix} \begin{pmatrix} c \\ d \end{pmatrix} = \mathbf{0} \quad (62)$$

Which, there are two special cases:

- First, if $\theta = 0$, then one gets the following:

$$\begin{pmatrix} 0 & 0 \\ 0 & -2 \end{pmatrix} \begin{pmatrix} a \\ b \end{pmatrix} = \mathbf{0} \implies v_1 = \begin{pmatrix} a \\ b \end{pmatrix} = \begin{pmatrix} a \\ 0 \end{pmatrix} \quad (63)$$

$$\begin{pmatrix} 2 & 0 \\ 0 & 0 \end{pmatrix} \begin{pmatrix} c \\ d \end{pmatrix} = \mathbf{0} \implies v_{-1} = \begin{pmatrix} c \\ d \end{pmatrix} = \begin{pmatrix} 0 \\ d \end{pmatrix} \quad (64)$$

So, one deduces that $\lambda = 1$ corresponds to normalized state $v_1 = \begin{pmatrix} 1 \\ 0 \end{pmatrix} = \chi_+$, and $\lambda = -1$ corresponds to normalized state $v_{-1} = \begin{pmatrix} 0 \\ 1 \end{pmatrix} = \chi_-$. Hence, after rescaling, $\pm \frac{\hbar}{2}$ corresponds to $\chi_{\pm}$ states correspondingly (which is consistent with the fact that $\theta = 0$ implies $\hat{r} = \hat{z}$).

- Then, if $\theta = \pi$, one gets the following instead:

$$\begin{pmatrix} -2 & 0 \\ 0 & 0 \end{pmatrix} \begin{pmatrix} a \\ b \end{pmatrix} = \mathbf{0} \implies v_1 = \begin{pmatrix} a \\ b \end{pmatrix} = \begin{pmatrix} 0 \\ b \end{pmatrix} \quad (65)$$

$$\begin{pmatrix} 0 & 0 \\ 0 & 2 \end{pmatrix} \begin{pmatrix} c \\ d \end{pmatrix} = \mathbf{0} \implies v_{-1} = \begin{pmatrix} c \\ d \end{pmatrix} = \begin{pmatrix} c \\ 0 \end{pmatrix} \quad (66)$$

So, one deduces that $\lambda = 1$ corresponds to normalized state $v_1 = \begin{pmatrix} 0 \\ 1 \end{pmatrix} = \chi_-$, and $\lambda = -1$ corresponds to normalized state $v_{-1} = \begin{pmatrix} 1 \\ 0 \end{pmatrix} = \chi_+$. Hence, after rescaling, $\pm \frac{\hbar}{2}$ corresponds to $\chi_{\mp}$ states correspondingly (which is consistent with the fact that $\theta = \pi$ implies $\hat{r} = -\hat{z}$, so the states flipped over).

For general case (where $\theta \in (0, \pi)$), since $\cos(\theta) - 1, \cos(\theta) + 1 \neq 0$, the equation becomes:

$$\begin{pmatrix} \cos(\theta) - 1 & \sin(\theta)e^{-i\phi} \\ \sin(\theta)e^{i\phi} & -\cos(\theta) - 1 \end{pmatrix} \begin{pmatrix} a \\ b \end{pmatrix} = \mathbf{0} \implies a + \frac{\sin(\theta)}{\cos(\theta) - 1} e^{-i\phi} b = 0 \quad (67)$$

$$\begin{pmatrix} \cos(\theta) + 1 & \sin(\theta)e^{-i\phi} \\ \sin(\theta)e^{i\phi} & -\cos(\theta) + 1 \end{pmatrix} \begin{pmatrix} c \\ d \end{pmatrix} = \mathbf{0} \implies c + \frac{\sin(\theta)}{\cos(\theta) + 1} e^{-i\phi} d = 0 \quad (68)$$

Which, take $b = \sqrt{\frac{1-\cos(\theta)}{2}} e^{i\phi} = \sin\left(\frac{\theta}{2}\right) e^{i\phi}$ and $d = \sqrt{\frac{1+\cos(\theta)}{2}} e^{i\phi} = \cos\left(\frac{\theta}{2}\right) e^{i\phi}$, one has the following:

$$a = -\frac{\sin(\theta)}{\cos(\theta) - 1} e^{-i\phi} b = -\frac{2 \sin(\theta/2) \cos(\theta/2)}{1 - \cos(\theta)} \sin(\theta/2) = \cos\left(\frac{\theta}{2}\right) \quad (69)$$

$$\implies |a|^2 + |b|^2 = \cos^2\left(\frac{\theta}{2}\right) + \sin^2\left(\frac{\theta}{2}\right) = 1 \quad (70)$$

$$c = -\frac{\sin(\theta)}{\cos(\theta) + 1} e^{-i\phi} d = -\frac{2 \sin(\theta/2) \cos(\theta/2)}{1 + \cos(\theta)} \cos(\theta/2) = -\sin\left(\frac{\theta}{2}\right) \quad (71)$$

$$\implies |c|^2 + |d|^2 = \sin^2\left(\frac{\theta}{2}\right) + \cos^2\left(\frac{\theta}{2}\right) = 1 \quad (72)$$

Hence, the states are normalized.

Passing through S_r , we get the eigenvalues and eigenvectors as follow:

$$\lambda = \frac{\hbar}{2} \implies v_1 = \begin{pmatrix} \cos(\theta/2) \\ \sin(\theta/2)e^{i\phi} \end{pmatrix} = \cos(\theta/2)\chi_+ + \sin(\theta/2)e^{i\phi}\chi_- \quad (73)$$

$$\lambda = -\frac{\hbar}{2} \implies v_{-1} = \begin{pmatrix} -\sin(\theta/2) \\ \cos(\theta/2)e^{i\phi} \end{pmatrix} = -\sin(\theta/2)\chi_+ + \cos(\theta/2)e^{i\phi}\chi_- \quad (74)$$

- (d) After measuring $\frac{\hbar}{2}$ for S_z , the state is at χ_+ . Which, notice the following relation of the eigenvectors deduced in part (c):

$$\cos(\theta/2)v_1 - \sin(\theta/2)v_{-1} = \chi_+ \quad (75)$$

Which, one if measure S_r now, the probability of getting $\frac{\hbar}{2}$, is the absolute value of coefficient square of v_1 (the corresponding eigenvector), which reads $\mathbb{P}(\lambda_r = \frac{\hbar}{2}) = |\cos(\theta/2)|^2 = \cos^2(\theta/2)$.

For $\theta = 0$ (the z -axis), one has this probability being 1; while $\theta = \frac{\pi}{2}$ (for the xy -plane, WLOG say it's the x -axis by choosing proper ϕ), it reads $\frac{1}{2}$, which coincides with the case of the spin in x -axis.

- (e) First, fix any specific orientation/Cartesian coordinates system. Given any state $\psi_r = a\chi_+ + b\chi_-$ (WLOG, we'll assume a is nonnegative real), there are two cases:

- First, if $a = 0$, then $\psi_r = b\chi_-$ with $|b|^2 = 1$, so it's just the eigenvalue of S_z , hence polarized in the default z -axis.
- Else, if $a \neq 0$, then since $a \in (0, 1]$ by assumption that $|a|^2 + |b|^2 = 1$ and a is nonnegative real, hence there exists $\theta \in [0, \pi)$, such that $a = \cos(\theta/2)$. As a result, one must have $|b|^2 = 1 - |a|^2 = 1 - \cos^2(\theta/2) = \sin^2(\theta/2)$, so $|b| = |\sin(\theta/2)|$, or $b = \sin(\theta)e^{i\phi}$ for some $\phi \in [0, 2\pi]$.

As a result, one has $\psi_r = \cos(\theta/2)\chi_+ + \sin(\theta)e^{i\phi}\chi_-$, which is precisely the up state v_1 for the axis $\hat{r} = \sin(\theta)\cos(\phi)\hat{x} + \sin(\theta)\sin(\phi)\hat{y} + \cos(\theta)\hat{z}$.

□

5 D

Question 5. *Electron in Oscillating Magnetic Field (Griffiths 4.36, 4.33 in second edition)*

An electron is at rest in an oscillating magnetic field $\mathbf{B} = B_0 \cos(wt)\hat{k}$, where B_0 and w are constants.

(a) Construct the Hamiltonian matrix for this system.

(b) The electron starts out (at $t = 0$) in the spin-up state with respect to the x -axis ($\chi(0) = \chi_+^{(x)}$). Determine $\chi(t)$ at any subsequent time.

Remark: This is a time-dependent Hamiltonian, so you cannot get $\chi(t)$ in the usual way from stationary states. Fortunately, in this case you can solve the time-dependent Schrödinger equation directly.

(c) Find the probability of getting $-\frac{\hbar}{2}$, if you measure S_x .

Answer: $\sin^2\left(\frac{\gamma B_0}{2w} \sin(wt)\right)$.

(d) What is the minimum field B_0 required to force a complete flip in S_x ?

Proof. (a) Given the hamiltonian $H = -\gamma \mathbf{B} \cdot \mathbf{S}$, let $\hat{k} = \hat{z}$, then the hamiltonian as a matrix is represented as follow:

$$\hat{H} = -\gamma B_0 \cos(wt) \mathcal{M}(S_z) = -\frac{\hbar \gamma B_0 \cos(wt)}{2} \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix} \quad (76)$$

(b) Given $\chi(t) = \begin{pmatrix} \chi_+(t) \\ \chi_-(t) \end{pmatrix}$, we have $\chi(0) = \chi_+^{(x)} = \frac{1}{\sqrt{2}} \begin{pmatrix} 1 \\ 1 \end{pmatrix}$ under the z -basis. If consider the time-dependent Schrödinger equation, one has the following:

$$i\hbar \frac{\partial}{\partial t} \begin{pmatrix} \chi_+(t) \\ \chi_-(t) \end{pmatrix} = \hat{H} \chi(t) = -\frac{\hbar \gamma B_0 \cos(wt)}{2} \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix} \begin{pmatrix} \chi_+(t) \\ \chi_-(t) \end{pmatrix} = -\frac{\hbar \gamma B_0 \cos(wt)}{2} \begin{pmatrix} \chi_+(t) \\ -\chi_-(t) \end{pmatrix} \quad (77)$$

Hence, one deduce the following differential equation:

$$i\hbar \frac{\partial \chi_+(t)}{\partial t} = -\frac{\hbar \gamma B_0 \cos(wt)}{2} \chi_+(t) \quad (78)$$

$$i\hbar \frac{\partial \chi_-(t)}{\partial t} = \frac{\hbar \gamma B_0 \cos(wt)}{2} \chi_-(t) \quad (79)$$

So, solving the equation, we get:

$$\frac{1}{\chi_+(t)} \frac{\partial \chi_+(t)}{\partial t} = \frac{i\gamma B_0}{2} \cos(wt) \implies \ln(\chi_+(t)) = \frac{i\gamma B_0}{2w} \sin(wt) + C \implies \chi_+(t) = A e^{\frac{i\gamma B_0}{2w} \sin(wt)} \quad (80)$$

$$\frac{1}{\chi_-(t)} \frac{\partial \chi_-(t)}{\partial t} = -\frac{i\gamma B_0}{2} \cos(wt) \implies \ln(\chi_-(t)) = -\frac{i\gamma B_0}{2w} \sin(wt) + D \implies \chi_-(t) = B e^{-\frac{i\gamma B_0}{2w} \sin(wt)} \quad (81)$$

Which, with the initial condition that $\chi_+(0) = \chi_-(0) = \frac{1}{\sqrt{2}}$, one has both $A, B = \frac{1}{\sqrt{2}}$. So, the state is given as follow:

$$\chi(t) = \frac{1}{\sqrt{2}} \begin{pmatrix} e^{\frac{i\gamma B_0}{2w} \sin(wt)} \\ e^{-\frac{i\gamma B_0}{2w} \sin(wt)} \end{pmatrix} \quad (82)$$

- (c) Given the above states, let's first rewrite it in the other form. Let $s(t) = \frac{\gamma B_0}{2w} \sin(wt)$, the state is given as follow:

$$\chi(t) = \frac{1}{\sqrt{2}} \begin{pmatrix} e^{is(t)} \\ e^{-is(t)} \end{pmatrix} = \frac{1}{\sqrt{2}} \begin{pmatrix} \cos(s(t)) + i \sin(s(t)) \\ \cos(s(t)) - i \sin(s(t)) \end{pmatrix} \quad (83)$$

$$= \cos(s(t)) \frac{1}{\sqrt{2}} \begin{pmatrix} 1 \\ 1 \end{pmatrix} + i \sin(s(t)) \frac{1}{\sqrt{2}} \begin{pmatrix} 1 \\ -1 \end{pmatrix} \quad (84)$$

$$= \cos(s(t)) \chi_+^{(x)} + i \sin(s(t)) \chi_-^{(x)} \quad (85)$$

Which, if measure S_x , the probability of getting $-\frac{\hbar}{2}$ is:

$$\mathbb{P}\left(\lambda_x = -\frac{\hbar}{2}\right) = |i \sin(s(t))|^2 = \sin^2\left(\frac{\gamma B_0}{2w} \sin(wt)\right) \quad (86)$$

- (d) To force a complete flip in S_x , since it's initially at $\chi_+^{(x)}$, one needs the coefficients of $\chi_-^{(x)}$ to have norm 1 at some point. Which, it's necessary for $|\sin(s(t))| = 1$ for some t , or $s(t) = \frac{\pi}{2} + 2k\pi$ for some $k \in \mathbb{Z}$, $t \geq 0$.

With $s(t) = \frac{\gamma B_0}{2w} \sin(wt)$, one has $-\frac{\gamma B_0}{2w} \leq s(t) \leq \frac{\gamma B_0}{2w}$ (while the bounds are minimum and maximum), so it's necessary for $\frac{\gamma B_0}{2w} \geq \frac{\pi}{2}$, then $B_0 \geq \frac{w\pi}{\gamma}$ is the minimum required magnetic field strength to cause a complete flip in S_x .

□